



Figure 3: Architecture of Big Data

- Hadoop:** This is the most admired execution of MapReduce, being a completely open source platform for handling Big Data. Hadoop is flexible enough to be able to work with many data sources, like aggregating data in order to do large scale processing, reading data from a database, etc.
- Hive:** This is an SQL-like link that allows BI applications to run queries beside a Hadoop cluster. Having been developed by Facebook, it has been made open source for a little while and is a higher-level concept of the Hadoop framework. Also, it allows everyone to make queries against data stored in a Hadoop cluster and has improved on Hadoop's functionality, making it ideal for BI users.

Advantages of Big Data processing

The capability of processing Big Data has various benefits.

- Businesses can make use of outside brainpower while taking decisions:** The right to use social data from search engines and websites like Facebook and Twitter is enabling enterprises to improve their business strategies.
- Enhanced customer service:** Customer response systems are getting replaced by new systems intended for Big Data technologies. Within these new systems, Big Data technologies are being utilised to read and assess consumer responses.
- Early recognition of risks for the services:** Risk factors can be recognised beforehand to deliver the perfect data.
- Improved operational competence:** Big Data technologies can be utilised for building staging areas or landing zones for new data, prior to deciding what

data should be moved to the data warehouse. Also, such incorporation of Big Data and data warehousing technologies helps businesses to bypass data that is not commonly accessed.

The challenges

Though it is very easy to get trapped in all the hype around Big Data, one of the reasons it is so underutilised is that there are many challenges still to be resolved in the technologies used to harness it. Some of these are:

- Companies face problems in identifying the correct data and examining how best to utilise it. Constructing data-related business cases frequently means forming opinions out-of-the-box and looking for income models that are extremely different from the traditional business model.
- Companies are reluctant to choose the fine talent that is capable of both working with new technologies and examining the data to find significant business insights.
- A bulk of data points have not been linked yet, and companies frequently do not have the correct platforms to combine and manage the data across the enterprise.
- The technology in the data world is evolving very fast. Leveraging data means functioning with well-built, pioneering technology collaborators – companies that can help create the right IT design so as to adapt to changes in the landscape in a well-organised manner.

The accessibility of Big Data, inexpensive product hardware, and new information managing and analytics software have come together to create a unique moment in the history of data analysis. We now have the capability that is necessary to examine these amazing data sets rapidly and cost-effectively, for the first time in history. This ability symbolises an authentic leap forward, and a chance to enjoy massive improvements in terms of work productivity, income and success. 

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